

TECHNICAL REPORT

Line Follower Car — Autonomous Mobile Robot

1. Introduction to the Mechanical Design

This technical report documents the complete mechanical design process of the Line Follower Car, an autonomous differential-drive mobile robot developed as a graduation/semester project for the Robotics course. The design was executed using Dassault Systèmes SolidWorks, and the physical prototypes were fabricated via Fused Deposition Modeling (FDM) 3D printing.

The mechanical architecture is based on a two-layer hierarchical chassis system — referred to throughout this document as the Upper Chassis (UC) and the Lower Chassis (LC). This dual-layer approach was deliberately selected to achieve the following engineering objectives:

- Orderly and systematic distribution of electronic and mechanical sub-assemblies across two independent structural planes. •
- A clear separation between actuation components (motors, wheels, caster) on the lower layer and control/sensing electronics (Arduino, Motor Driver, Batteries) on the upper layer. •
- Facilitation of maintenance, debugging, and component replacement without full disassembly. •
- Optimization of the robot's Center of Gravity (CoG) by positioning high-mass items (batteries) in a low central position. •

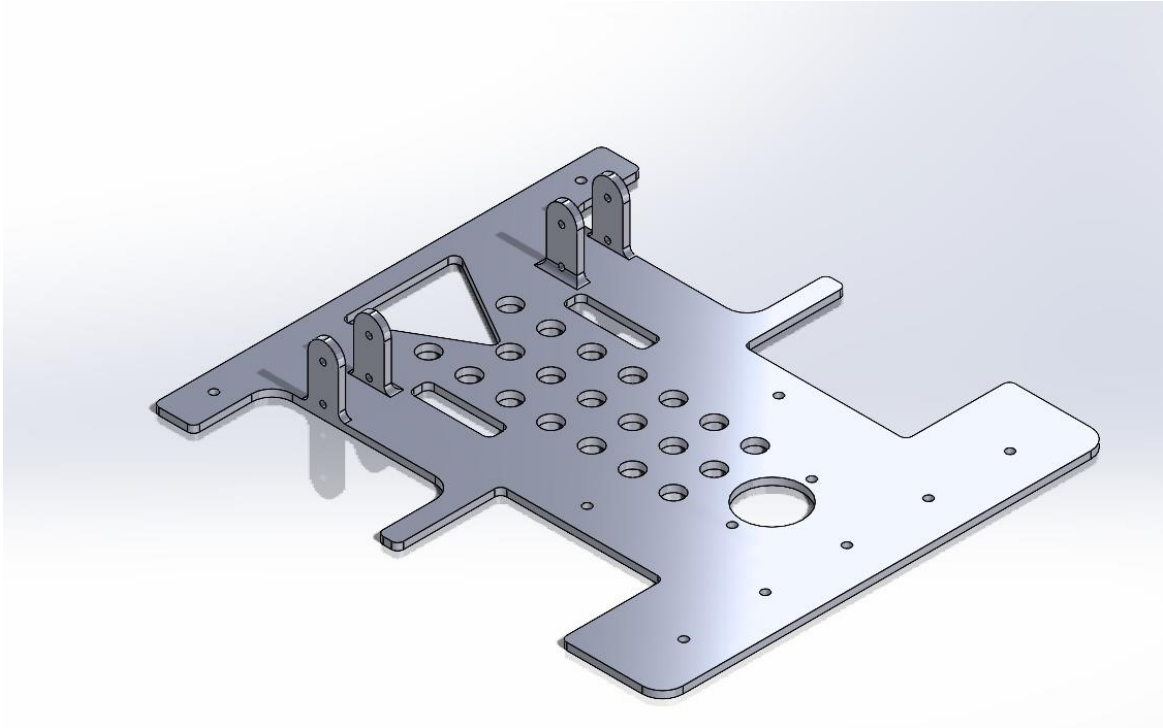


Figure 1: Isometric view of the complete Line Follower Car assembly.

2. Design Methodology (Design Approach)

2.1 Phase 1: Physical Component Measurement

Prior to initiating any CAD modeling in SolidWorks, a rigorous physical measurement phase was conducted on all actual hardware components. Each component was measured using a digital vernier caliper and cross-referenced against manufacturer datasheets to validate dimensional accuracy. The measured dimensions were then used as the primary input parameters for the SolidWorks sketches.

The following key component dimensions were collected and verified:

Component	Model / Type	Key Dimensions	Quantity
DC Geared Motor + Wheel	Dual Shaft 3:12VDC, 600 RPM	65×22×19 mm (motor); Ø65mm (wheel)	2
Motor Driver	L298N Module	43×43×27 mm	1
Arduino Uno	Rev3	68.6×53.4 mm	1
Battery Holder (2×18650)	2-Slot Series	76×40×19 mm	1
Metal Caster Wheel	Steel Ball Caster	48×40×20 mm	1
IR Line Tracker Sensor	TCRT5000 Module	31×14 mm (board)	6
IR Photoelectric Encoder Sensor	4-Pin Slotted Optical	31×13.3 mm	2
Encoder Wheel	20-Slot, 26mm diameter	Ø26 mm	2

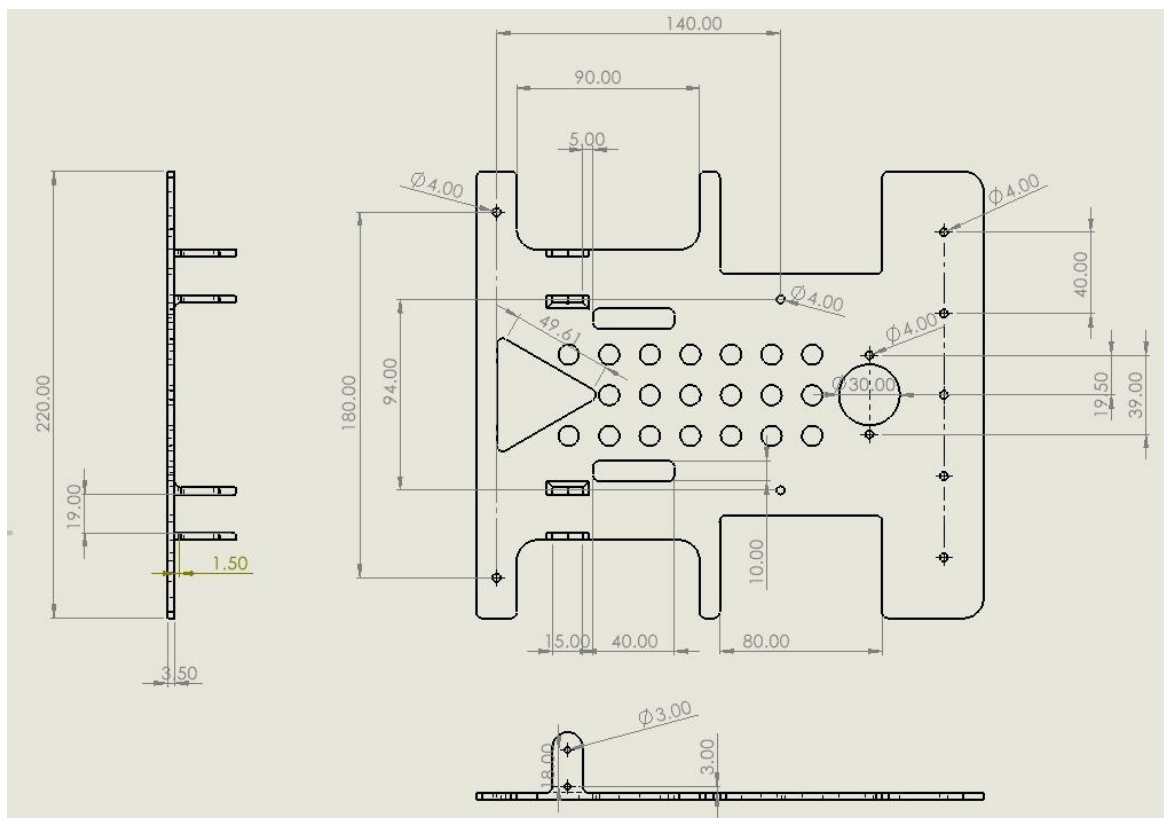
Breadboard	830-Point, Split Rails	165×55 mm	1
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Table 1: Hardware component inventory with measured/specified dimensions.

2.2 Phase 2: Weight Distribution & Stability Analysis

A fundamental constraint in differential-drive robot design is achieving a stable and balanced Center of Gravity (CoG). An improperly balanced CoG results in unequal normal forces on the drive wheels, leading to unpredictable heading drift during line-following operation. The following distribution strategy was implemented:

- The two 18650 lithium-ion batteries (the highest-mass components) were positioned centrally on the upper chassis, directly above the axle centerline, to minimize the moment arm of gravitational force relative to the drive axis.
- The Arduino Uno and Motor Driver module were placed on the upper chassis in a symmetric, balanced arrangement relative to the robot's longitudinal axis.
- The metal caster wheel (front omnidirectional support) was positioned at the front underside of the lower chassis to form a stable three-point contact triangle with the two drive wheels.
- The drive motors were symmetrically placed on the left and right sides of the lower chassis, ensuring equal mechanical leverage about the robot's center.



3. Chassis Technical Specifications

3.1 Overall Dimensions

The overall dimensions of the chassis assembly were determined to provide adequate mounting area for all components while maintaining a compact footprint suitable for standard line-following track widths.

Parameter	Value
Overall Robot Length	320 mm (32 cm)
Overall Robot Width	220 mm (22 cm)
Chassis Plate Thickness (LC & UC)	3 mm (PLA / FDM Printed)
Inter-Layer Standoff Height	Defined by 70 mm standoff screws
Ground Clearance	~19 mm (caster wheel height)
Drive Wheel Diameter	65 mm
Wheelbase (motor-to-motor)	~140 mm
Material	PLA (Polylactic Acid) — FDM 3D Printed
Dimensional Tolerance	±0.5 mm

Table 2: Principal dimensional parameters of the chassis assembly.

3.2 Hole Pattern Philosophy

The hole placement strategy employed in both the Upper Chassis (UC) and Lower Chassis (LC) was not arbitrary; each aperture serves a specific structural or functional purpose as described below:

3.2.1 Inter-Layer Fastening Holes

Four M3 clearance holes ($\varnothing$ 4 mm) are located at the four corners of both the UC and LC. These holes accommodate M3×70mm hex standoff bolts (with nuts) that rigidly connect the two chassis layers and maintain a fixed inter-layer separation. The corner positions were chosen to maximize the structural stiffness of the two-layer assembly.

3.2.2 Motor Mounting Holes

Dedicated $\varnothing$ 3 mm holes are provided on the lateral protrusions (motor bay cutouts) of the LC. The DC Geared Motors are secured to the chassis via their integral mounting bracket, using M3×10mm bolts and hex nuts. The hole positions were calculated based on the motor's mounting-hole center-to-center distance as extracted from the manufacturer's mechanical drawing.

3.2.3 Caster Wheel Mounting Holes

A pair of $\varnothing$ 3 mm holes (spaced 32 mm center-to-center along the longitudinal axis) accommodates the metal caster wheel mounting plate at the front underside of the LC. The caster centerline was aligned precisely with the robot's longitudinal axis of symmetry to prevent passive yaw bias.

3.2.4 Encoder Sensor Mounting Holes

Two $\varnothing$ 3 mm holes per side (total of four holes) are positioned adjacent to the motor bays on the LC. These holes accept M3 machine screws that clamp the IR Photoelectric Encoder Sensor

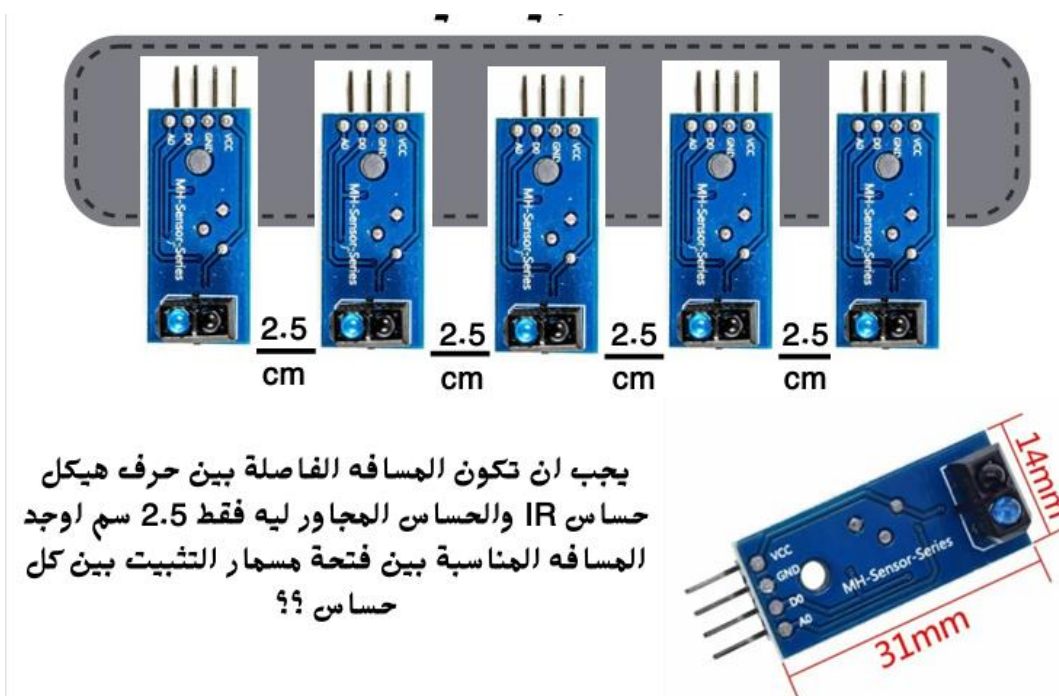
modules in position, aligning the sensor's optical gap with the mid-plane of the 20-slot encoder wheel mounted on the motor's secondary output shaft.

3.2.5 Wire Routing Apertures

Rectangular slot cutouts (dimensions: 15 mm × 8 mm) are strategically located in both chassis plates. These slots serve as wire management passages, allowing motor cables, sensor harnesses, and power leads to pass between the upper and lower layers in an organized manner, preventing cable entanglement with moving parts (wheels, encoder wheels).

3.2.6 IR Sensor Array Mounting Slots

A dedicated mounting rail zone at the front edge of the LC accommodates the six TCRT5000 IR line-tracking sensors. Each sensor is separated by a 25 mm center-to-center pitch, providing a total array span of 125 mm. The mounting holes accept M2 screws to secure the sensor PCBs at a flush, downward-facing orientation approximately 3–8 mm above the floor surface.



3.2.7 Electronic Component Mounting Holes

The UC features a grid of Ø3 mm holes matching the mounting hole patterns of the Arduino Uno (68.6×53.4 mm board outline, 4 mounting holes), the L298N Motor Driver (43×43 mm, 4 mounting holes), and a dedicated attachment region for the breadboard (165×55 mm, secured with M2.5 screws through the PCB mounting rails).

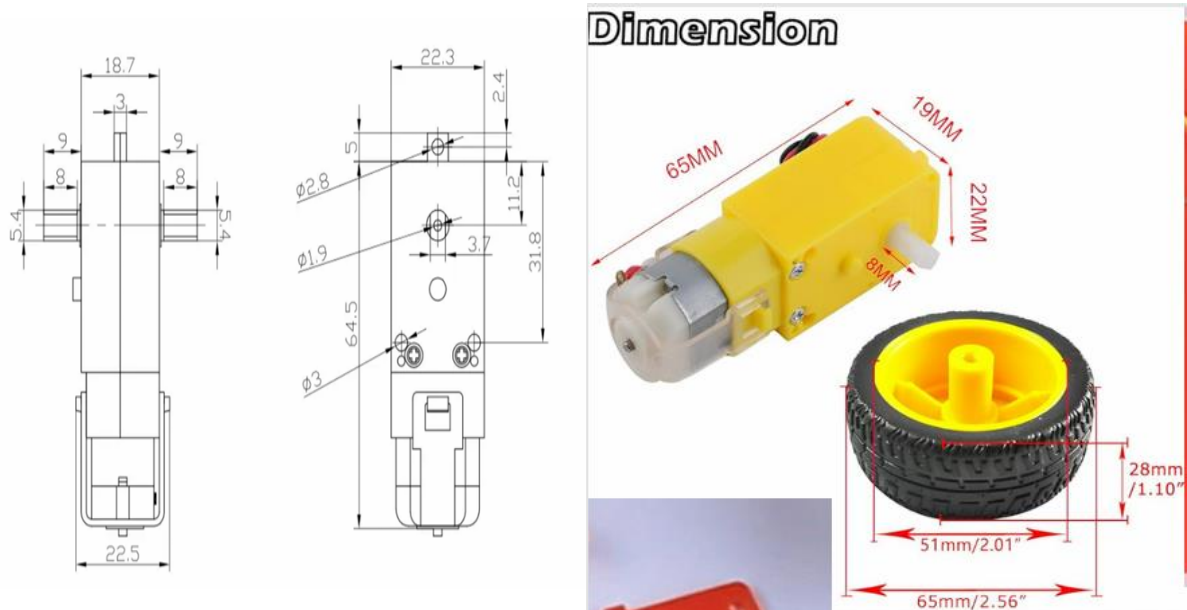
4. Component Placement & Layout

4.1 Lower Chassis (LC) — Actuation Layer

The LC serves as the primary actuation and mobility platform. All components directly involved in the robot's locomotion are mounted on or through this layer.

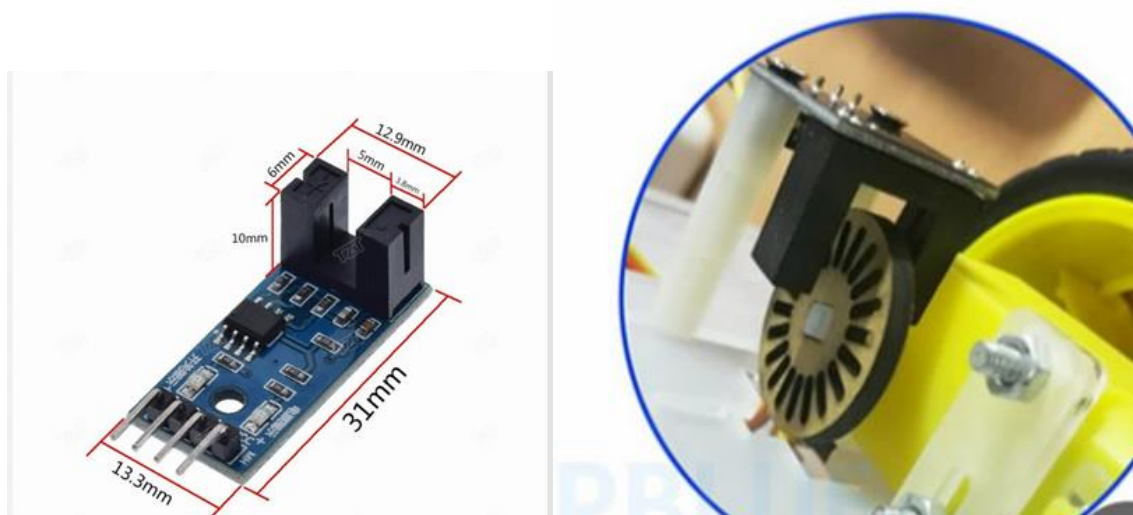
4.1.1 DC Geared Motors & Drive Wheels

Two DC Geared Motor modules (each with an integrated rubber wheel, Ø65 mm × W26 mm) are mounted symmetrically on the left and right lateral bays of the LC. The motor axis is positioned at the longitudinal midpoint of the chassis to balance the robot's moment of inertia. The motor brackets protrude laterally beyond the chassis outline to align the wheel treads flush with the robot's outermost envelope, ensuring straight-line tracking. The motor shafts are oriented horizontally, perpendicular to the robot's forward direction.



4.1.2 Encoder Wheels & Sensors

Each DC Motor's secondary (rear-facing) shaft carries a 20-slot encoder wheel (Ø26 mm). The corresponding IR Photoelectric Encoder Sensor modules are mounted on the LC, with their optical slot aligned to straddle the encoder wheel's perimeter. The mounting holes were calculated to position the sensor's optical gap at exactly the midpoint of the encoder wheel thickness (± 0.5 mm tolerance). This arrangement provides 20 pulses per revolution, enabling closed-loop speed feedback to the Arduino.



4.1.3 Metal Caster Wheel

The metal ball caster wheel (flange base: 48 mm × 40 mm, ball diameter: 15 mm, total height: 20 mm) is mounted at the front-center underside of the LC. The caster's ball center is positioned on the robot's longitudinal axis of symmetry ($X = 0$), at a distance from the drive wheel axle calculated to ensure that the chassis floor remains parallel to the ground surface. The caster is secured with 2× M3 machine screws through the flange mounting holes.

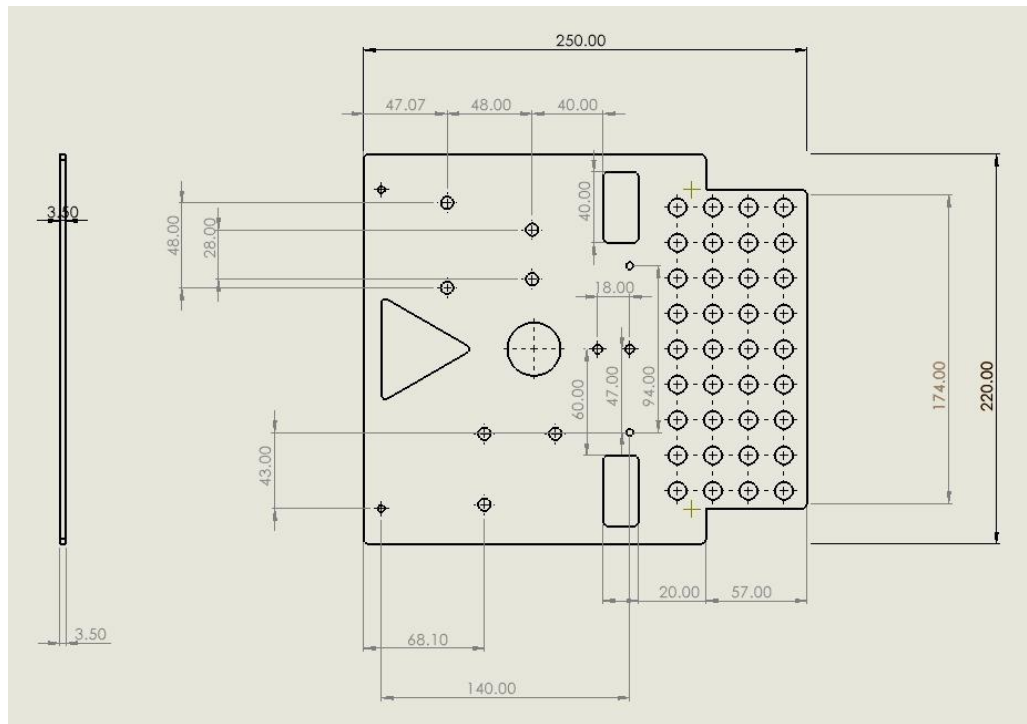


4.1.4 IR Sensor Array (Line Tracking)

Six TCRT5000 IR Line Tracker sensor modules are mounted in a horizontal linear array across the front edge of the LC. The array is configured with a 25 mm uniform pitch between sensor centers, providing a total sensing width of 125 mm. Each sensor is oriented face-down with the emitter/detector pair pointing toward the floor at a working height of 5–8 mm. The central sensor is aligned with the robot's longitudinal axis. This configuration provides sufficient lateral coverage to detect line deviations and enable robust PID-based correction.

4.2 Upper Chassis (UC) — Control & Power Layer

The UC provides a clean, elevated mounting surface for all electronic control and power components, keeping them separated from the mechanical noise (vibration, motor EMI) of the actuation layer.

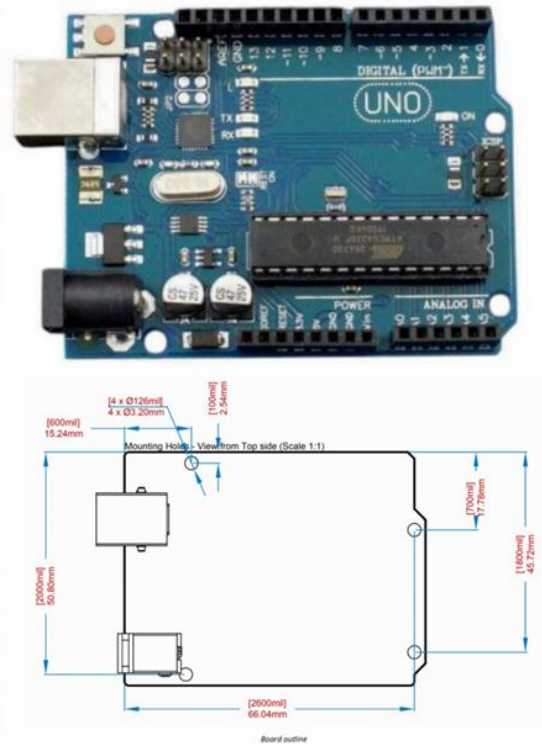
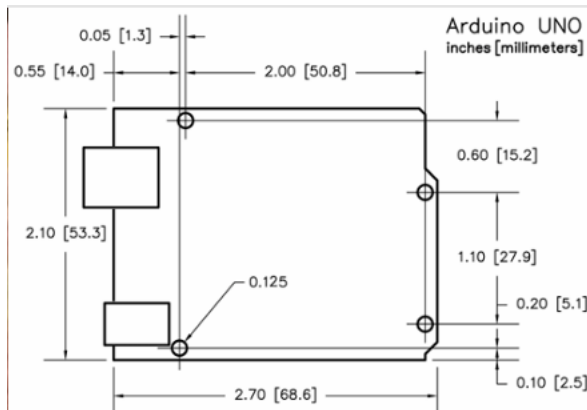


4.2.1 Battery Placement (CoG Optimization)

The 2× 18650 lithium-ion cell holder (76×40×19 mm) is mounted at the geometric center of the UC, directly above the drive wheel axle centerline. This placement was a deliberate design decision based on two criteria: (1) minimizing the longitudinal CoG offset from the drive axis, and (2) maintaining a low CoG height to improve roll stability on uneven surfaces. The battery holder is secured with M3×10 mm bolts through matching holes in the UC.

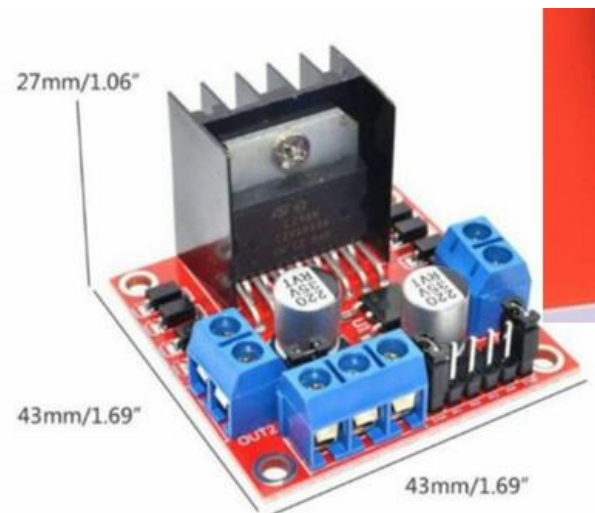
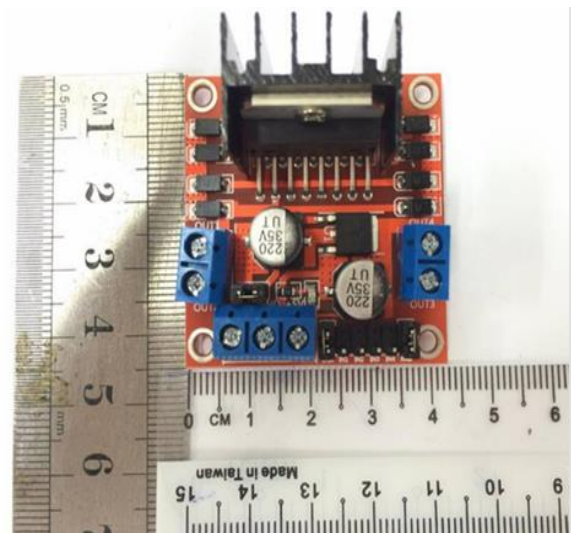
4.2.2 Arduino Uno

The Arduino Uno (66×53.4 mm PCB) is mounted on the left half of the UC, secured via its four M3 mounting holes. The board is oriented with its USB-B programming port and power jack accessible from the robot's rear face for easy field programming and battery connection. The digital I/O headers face inward toward the breadboard to minimize jumper wire lengths.



4.2.3 L298N Motor Driver

The L298N Motor Driver module (43×43 mm) is mounted on the right side of the UC, adjacent to the motor cable entry aperture. This proximity to the motor cables minimizes the length of high-current wiring between the driver and the motors, reducing resistive losses and electromagnetic emissions. The module's screw-terminal connectors face the motor cable side for clean wire management.



4.2.4 Breadboard

The 830-point breadboard (165×55 mm) is mounted longitudinally along the rear portion of the UC. Its placement adjacent to the Arduino's GPIO headers allows short Male-to-Male jumper connections. An important design modification was made to the UC to accommodate the breadboard's full footprint: the UC width in the breadboard zone was extended by 80 mm relative to the initial design iteration.



5. Design Optimization & Cost Reduction

5.1 Challenge: Excessive Print Mass in Initial Design

The initial SolidWorks design iteration produced solid rectangular chassis plates (3 mm thickness) without any internal material removal. When this geometry was imported into the 3D printing slicer software, the estimated print mass and material consumption were found to be significantly high. Specifically, the combination of both chassis plates would have consumed an excessive volume of PLA filament, leading to two primary problems:

- Excessive print cost: High PLA filament consumption translates directly to increased fabrication cost, which was inconsistent with the project's material budget. •
- Excessive print time: Large solid areas significantly increase slicer-estimated print time, which was impractical given the project timeline. •
- Unnecessary mass: A heavier chassis requires higher motor torque to achieve the same acceleration, potentially exceeding the stall torque of the selected 600 RPM DC Geared Motors. •

5.2 Solution: Geometric Lightweighting (Pocketing)

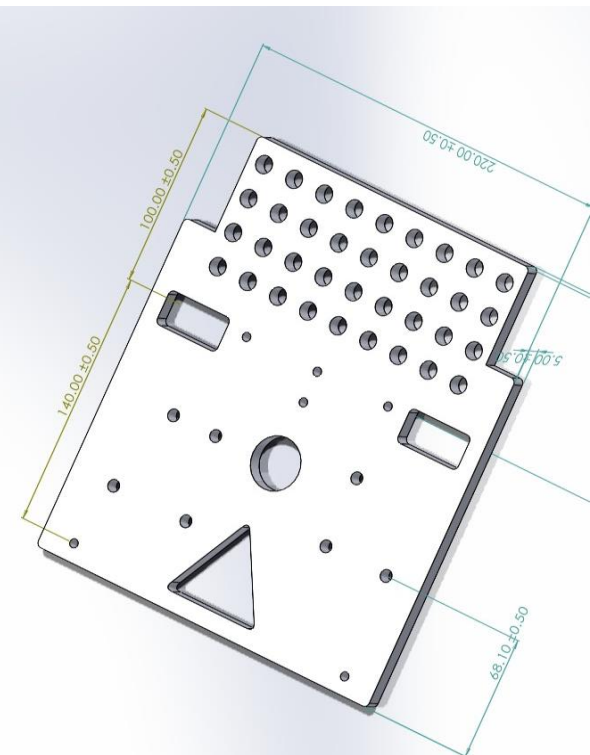
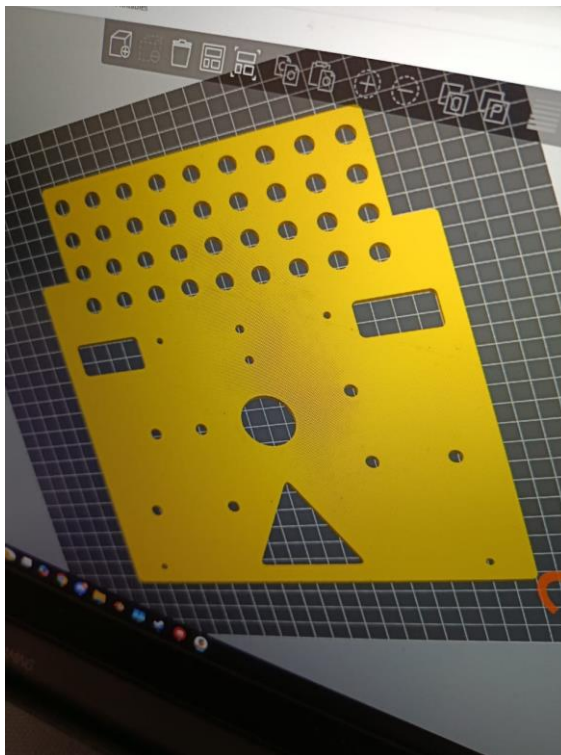
A systematic lightweighting strategy was implemented in SolidWorks by introducing material-removal pockets in the non-load-bearing zones of both chassis plates. The pocketing geometry was designed to satisfy two simultaneous objectives: functional (structural integrity) and aesthetic (visual quality of the final robot).

The following geometric features were incorporated:

- Triangular cutouts: Triangular pockets were introduced in the central body of both the UC and LC. Triangles are structurally efficient shapes (inherently rigid under in-plane loading) and provide a recognizable visual identity to the design. •

- Circular perforations: A grid pattern of circular holes ($\varnothing 8\text{--}10\text{ mm}$) was applied across the non-critical flat surface areas of both chassis plates, reducing plate mass without compromising bending stiffness.
- Rectangular slots: Elongated rectangular apertures were introduced in zones where only wire routing or air circulation was needed, serving a dual functional-aesthetic purpose.

The weight reduction achieved through pocketing was estimated at approximately 35–45% of the original solid-plate mass, with no measurable reduction in the structural rigidity of the chassis under the applied load case (distributed component weight + dynamic loads from motor vibration). This was validated qualitatively by the SolidWorks static simulation results.



5.3 Structural Rigidity Verification

The following qualitative criteria were used to confirm that the lightweighting did not compromise structural performance:

- Load path continuity: Pockets were placed only in areas where no primary load transfer occurs between fastener points.
- Minimum web thickness: All remaining material bridges between pockets maintain a minimum width of 8 mm to prevent stress concentration-induced crack initiation.
- Critical mounting zones preserved: The areas immediately surrounding all M3 fastener holes were kept solid (no pocketing within 5 mm of any hole center) to ensure adequate bearing strength.

6. Assembly Procedure

6.1 Required Hardware

Fastener Type	Size	Quantity
Machine Screw (motor & sensor mounting)	M3 × 70 mm	4
Machine Screw (caster, Arduino, driver)	M3 × 20 mm	6
Machine Screw (sensor, encoder, PCB)	M3 × 10 mm	18
Hex Nut	M3 (standard)	35

Table 3: Fastener inventory for the assembly procedure.

6.2 Step-by-Step Assembly Sequence

1. LOWER CHASSIS (LC) PREPARATION: Place the 3D-printed LC flat on a clean surface, bottom face up. Verify that all hole diameters are clear of print artifacts (use a 3 mm drill if needed to clear tight holes).
2. MOTOR INSTALLATION: Insert each DC Geared Motor into its lateral bay cutout from the inside face of the LC. Align the motor bracket holes with the Ø3 mm mounting holes. Secure with M3×10 mm screws and hex nuts, tightening in a cross-pattern. Repeat for both motors. Verify that wheel rotation is unobstructed.
3. CASTER WHEEL INSTALLATION: Mount the metal caster wheel on the front underside of the LC, aligning the caster flange holes with the Ø3 mm caster mounting holes. Secure with 2× M3×10 mm screws and hex nuts. Verify the caster ball rolls freely in all directions.
4. ENCODER WHEEL INSTALLATION: Press-fit the 20-slot encoder wheel onto the secondary shaft of each motor. Ensure the encoder wheel is fully seated and perpendicular to the shaft axis (check by eye against the chassis face).
5. ENCODER SENSOR INSTALLATION: Align each IR Encoder Sensor module adjacent to its corresponding motor, positioning the optical slot around the perimeter of the encoder wheel. Secure with 2× M3×10 mm screws. Verify the encoder wheel rotates freely through the optical gap without contact.
6. IR LINE SENSOR ARRAY INSTALLATION: Secure the six TCRT5000 IR sensor modules in the front sensor rail. Mount each sensor PCB with the emitter/detector face pointing downward. Space sensors at 25 mm pitch. Use M2.5 screws where available, or adhesive mounts for the TCRT5000 boards.
7. WIRE ROUTING (LOWER LAYER): Route all motor cables and sensor wires through the designated rectangular wire routing slots in the LC. Bundle wires with cable ties to prevent interference with wheels and encoder wheels.
8. UPPER CHASSIS (UC) PREPARATION: Place the UC on a flat surface. Attach M3×70 mm hex standoff pillars to the four corner fastening holes of the LC, using hex nuts on the underside of the LC.
9. BATTERY HOLDER INSTALLATION (UC): Mount the 2×18650 battery holder at the central position on the UC using M3×10 mm screws. Connect the battery holder output leads (red = V+, black = GND) as per the circuit wiring diagram.
10. ARDUINO UNO INSTALLATION (UC): Align the Arduino Uno PCB over its designated mounting holes on the UC. Secure with M3×10 mm screws through the Arduino's four mounting holes with hex nuts underneath.
11. L298N MOTOR DRIVER INSTALLATION (UC): Mount the L298N Motor Driver module on the right-side zone of the UC. Secure with M3×10 mm screws. Connect motor output terminals (OUT1–OUT4) to the corresponding motor leads routed up through the LC wire slots.
12. BREADBOARD INSTALLATION (UC): Attach the 830-point breadboard to the UC using the breadboard's self-adhesive backing or M2.5 screws through the PCB rail. Ensure the breadboard's power rails (red/blue) are oriented toward the Arduino's 5V and GND pins.

TWO-LAYER ASSEMBLY: Lower the UC onto the standoff pillars protruding from the LC corners. Secure the UC with M3 hex nuts on top of each standoff. Apply 0.5–1.0 N·m torque (hand-tight plus 1/4 turn).

FINAL WIRING & VERIFICATION: Complete all jumper wire connections per the circuit schematic. Verify continuity for all signal and power connections before applying power. Conduct a free-wheel spin test of both drive wheels and the caster wheel to confirm no mechanical interference.

7. Conclusion

The mechanical design of the Line Follower Car successfully fulfills all functional and structural requirements identified during the design specification phase. The two-layer hierarchical chassis architecture achieves a clear separation of concerns between the actuation domain (Lower Chassis) and the electronics domain (Upper Chassis), resulting in a mechanically organized, maintainable, and aesthetically refined robotic platform.

The design-for-manufacturing approach — beginning with physical component measurement and proceeding through weight distribution analysis, hole pattern design, and lightweighting optimization — demonstrates a disciplined engineering methodology applicable to real-world product development. The pocketing strategy reduced print cost and mass by an estimated 35–45% while maintaining full structural rigidity under operational loading.

The final chassis dimensions of 320 mm (L) × 220 mm (W) × 3 mm (t) provide adequate component footprint with a competitive overall robot envelope. The design is ready for seamless integration with the electronics, sensor, and control software developed by the other team members.

8. References & Component Datasheets

- DC Geared Motor Dual Shaft 3:12VDC 600 RPM — Manufacturer Mechanical Drawing •
- Arduino Uno Rev3 — Official Dimensional Drawing, [arduino.cc](https://www.arduino.cc) •
- L298N Motor Driver Module — Datasheet, ST Microelectronics •
- TCRT5000 IR Reflective Sensor — Datasheet, Vishay Semiconductors •
- IR Photoelectric Encoder Sensor (FC-33 compatible) — Module Dimensional Specifications •
- Metal Caster Wheel for Robot — Product Dimensional Sheet (48×40×20 mm) •
- Encoder Wheel, 20-Slot, 26mm — Product Specifications •
- Dassault Systèmes SolidWorks 2024 — CAD Software Platform •
- FDM 3D Printing Guidelines — PLA Material Properties & Tolerances •